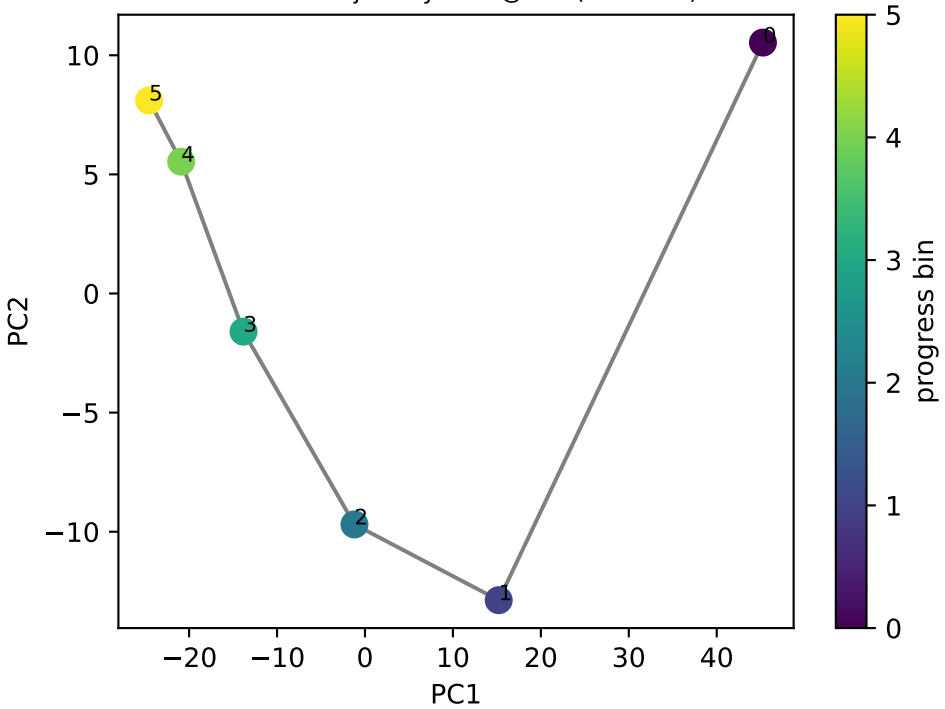
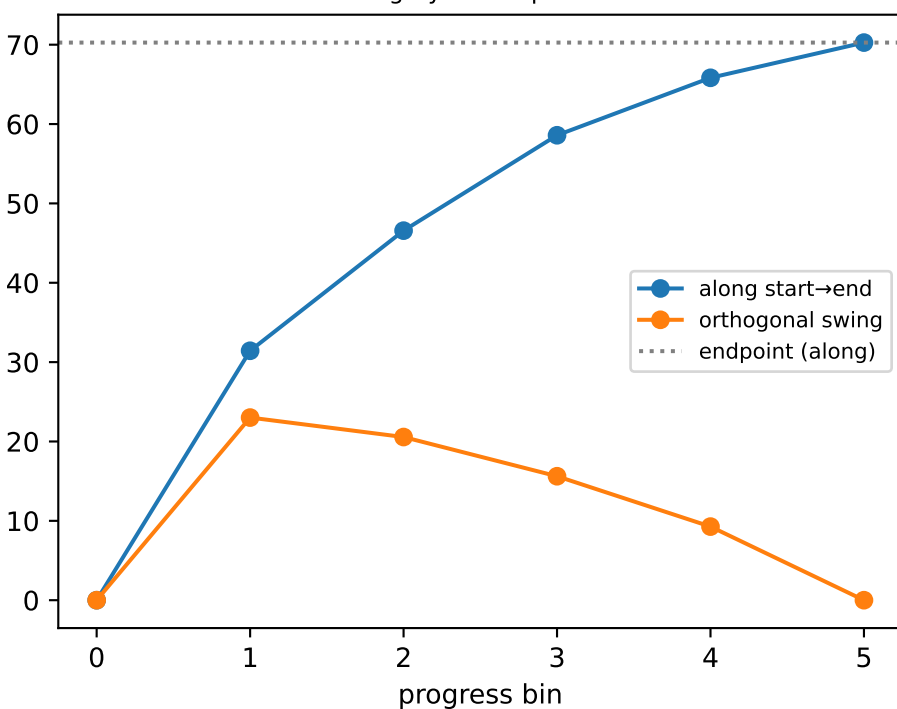


Qwen3-32B convergence trajectory — turn-resolved manifold (equal per-seed, normalized progress)

centroid trajectory PCA @L48 (PR=1.39)



swing-by decomposition



speed & curvature

